

|    | A                                      | B        | C            | D             | E                | F          | G          | H             | I      | J           | K          | L          | M        | N                       | O        | P    | Q      | R       | U          | V          | W                 | X        | Y       | Z | AA |
|----|----------------------------------------|----------|--------------|---------------|------------------|------------|------------|---------------|--------|-------------|------------|------------|----------|-------------------------|----------|------|--------|---------|------------|------------|-------------------|----------|---------|---|----|
| 1  | 365 DataScience RE California Database |          |              |               |                  |            |            |               |        |             |            |            |          |                         |          |      |        |         |            |            |                   |          |         |   |    |
| 2  | Practical example                      |          |              |               |                  |            |            |               |        |             |            |            |          |                         |          |      |        |         |            |            |                   |          |         |   |    |
| 3  |                                        |          |              |               |                  |            |            |               |        |             |            |            |          |                         |          |      |        |         |            |            |                   |          |         |   |    |
| 4  | Product                                |          |              |               |                  |            |            |               |        |             | Customer   |            |          |                         |          |      |        |         |            |            |                   |          |         |   |    |
| 5  | ID                                     | Building | Year of sale | Month of sale | Type of property | Property # | Area (ft.) | Price         | Status | Customer ID | Entity     | Name       | Surname  | Age at time of purchase | Interval | Y    | Gender | Country | State      | Purpose    | Deal satisfaction | Mortgage | Source  |   |    |
| 6  | 1030                                   | 1        | 2005         |               | 11 Apartment     | 30         | 743.09     | \$ 246,172.68 | Sold   | C0028       | Individual | Madalyn    | Mercer   | 19                      | 18-25    | 1986 | F      | USA     | California | Home       | 5 No              |          | Website |   |    |
| 7  | 1029                                   | 1        | 2005         |               | 10 Apartment     | 29         | 756.21     | \$ 246,331.90 | Sold   | C0027       | Individual | Lara       | Carrillo | 22                      | 18-25    | 1983 | F      | USA     | California | Home       | 5 No              |          | Website |   |    |
| 8  | 2002                                   | 2        | 2007         |               | 7 Apartment      | 2          | 587.28     | \$ 209,280.91 | Sold   | C0112       | Individual | Donavan    | Flowers  | 22                      | 18-25    | 1985 | M      | USA     | California | Home       | 1 Yes             |          | Client  |   |    |
| 9  | 2031                                   | 2        | 2007         |               | 12 Apartment     | 31         | 1604.75    | \$ 452,667.01 | Sold   | C0160       | Individual | Darien     | Dorsey   | 22                      | 18 25    | 1985 | M      | USA     | California | Investment | 3 Yes             |          | Website |   |    |
| 10 | 1049                                   | 1        | 2004         |               | 11 Apartment     | 49         | 1375.45    | \$ 467,083.31 | Sold   | C0014       | Individual | Alessandra | Perry    | 25                      | 18-25    | 1979 | F      | USA     | California | Home       | 4 No              |          | Agency  |   |    |
| 11 | 3011                                   | 3        | 2007         |               | 9 Apartment      | 11         | 675.19     | \$ 203,491.85 | Sold   | C0125       | Individual | Kaitlin    | Owen     | 26                      | 26-35    | 1981 | F      | USA     | Virginia   | Investment | 5 No              |          | Client  |   |    |
| 12 | 3026                                   | 3        | 2007         |               | 9 Apartment      | 26         | 670.89     | \$ 212,520.83 | Sold   | C0125       | Individual | Kaitlin    | Owen     | 26                      | 26-35    | 1981 | F      | USA     | Virginia   | Investment | 5 No              |          | Agency  |   |    |
| 13 | 3023                                   | 3        | 2008         |               | 1 Apartment      | 23         | 720.81     | \$ 198,591.85 | Sold   | C0166       | Individual | Terry      | Forbes   | 26                      | 26-35    | 1982 | M      | USA     | California | Home       | 5 No              |          | Client  |   |    |
| 14 | 1031                                   | 1        | 2006         |               | 6 Apartment      | 31         | 782.25     | \$ 265,467.68 | Sold   | C0034       | Individual | Kole       | Shannon  | 27                      | 26-35    | 1979 | M      | USA     | Arizona    | Home       | 2 Yes             |          | Website |   |    |
| 15 | 4023                                   | 4        | 2006         |               | 3 Apartment      | 23         | 794.52     | \$ 235,633.26 | Sold   | C0170       | Individual | Emmy       | Singh    | 27                      | 26-35    | 1979 | F      | USA     | Virginia   | Investment | 3 Yes             |          | Agency  |   |    |
| 16 | 1036                                   | 1        | 2004         |               | 10 Apartment     | 36         | 1160.36    | \$ 317,473.86 | Sold   | C0009       | Individual | Arabella   | Ferrell  | 28                      | 26-35    | 1976 | F      | USA     | Oregon     | Home       | 1 No              |          | Agency  |   |    |
| 17 | 1046                                   | 1        | 2006         |               | 8 Apartment      | 46         | 1942.50    | \$ 503,790.23 | Sold   | C0041       | Individual | Christian  | Costa    | 26                      | 26-35    | 1980 | M      | USA     | California | Home       | 5 No              |          | Website |   |    |
| 18 | 4035                                   | 4        | 2007         |               | 10 Apartment     | 35         | 794.52     | \$ 217,786.38 | Sold   | C0057       | Individual | Michelle   | Cameron  | 29                      | 26-35    | 1978 | F      | USA     | Nevada     | Home       | 3 Yes             |          | Website |   |    |
| 19 | 2036                                   | 2        | 2006         |               | 11 Apartment     | 36         | 1109.25    | \$ 460,001.26 | Sold   | C0061       | Individual | Enrique    | Cardenas | 29                      | 26-35    | 1977 | M      | USA     | California | Home       | 2 No              |          | Website |   |    |
| 20 | 2056                                   | 2        | 2007         |               | 4 Apartment      | 56         | 1400.95    | \$ 460,001.26 | Sold   | C0089       | Individual | Amanda     | Simon    | 29                      | 26-35    | 1978 | F      | USA     | California | Home       | 5 No              |          | Agency  |   |    |
| 21 | 1047                                   | 1        | 2007         |               | 12 Apartment     | 47         | 1479.72    | \$ 448,134.27 | Sold   | C0159       | Individual | Kamden     | Stewart  | 29                      | 26-35    | 1978 | M      | USA     | California | Home       | 5 No              |          | Website |   |    |
| 22 | 5051                                   | 5        | 2006         |               | 3 Apartment      | 51         | 790.54     | \$ 249,591.99 | Sold   | C0171       | Individual | Skylar     | Buchanan | 29                      | 26-35    | 1977 | M      | USA     | Nevada     | Home       | 4 Yes             |          | Website |   |    |
| 23 | 2007                                   | 2        | 2006         |               | 8 Apartment      | 7          | 723.93     | \$ 196,142.19 | Sold   | C0042       | Individual | Michael    | Juarez   | 30                      | 26-35    | 1976 | M      | USA     | Colorado   | Home       | 4 Yes             |          | Agency  |   |    |
| 24 | 3020                                   | 3        | 2007         |               | 4 Apartment      | 20         | 781.07     | \$ 258,572.48 | Sold   | C0093       | Individual | Antonio    | Porter   | 30                      | 26-35    | 1977 | M      | USA     | Arizona    | Home       | 5 No              |          | Client  |   |    |
| 25 | 3029                                   | 3        | 2007         |               | 4 Apartment      | 29         | 1127.76    | \$ 310,831.21 | Sold   | C0093       | Individual | Antonio    | Porter   | 30                      | 26-35    | 1977 | M      | USA     | Arizona    | Home       | 3 No              |          | Client  |   |    |
| 26 | 3015                                   | 3        | 2006         |               | 10 Apartment     | 15         | 720.70     | \$ 207,281.59 | Sold   | C0051       | Individual | Conner     | Huff     | 31                      | 26-35    | 1975 | M      | USA     | Nevada     | Home       | 5 No              |          | Website |   |    |
| 27 | 2004                                   | 2        | 2006         |               | 12 Apartment     | 4          | 649.69     | \$ 168,834.04 | Sold   | C0064       | Individual | Joaquin    | Mullins  | 31                      | 26-35    | 1975 | M      | USA     | California | Investment | 5 No              |          | Agency  |   |    |
| 28 | 2006                                   | 2        | 2006         |               | 12 Apartment     | 6          | 1307.45    | \$ 396,973.83 | Sold   | C0064       | Individual | Joaquin    | Mullins  | 31                      | 26-35    | 1975 | M      | USA     | California | Investment | 5 No              |          | Agency  |   |    |
| 29 | 5013                                   | 5        | 2007         |               | 9 Apartment      | 13         | 618.38     | \$ 188,743.11 | Sold   | C0128       | Individual | Kyla       | Walker   | 31                      | 26-35    | 1976 | F      | USA     | Colorado   | Home       | 4 Yes             |          | Agency  |   |    |
| 30 | 1026                                   | 1        | 2005         |               | 3 Apartment      | 26         | 625.80     | \$ 179,674.08 | Sold   | C0019       | Individual | Victor     | Jensen   | 32                      | 26-35    | 1973 | M      | USA     | California | Home       | 4 No              |          | Website |   |    |
| 31 | 2054                                   | 2        | 2006         |               | 6 Apartment      | 54         | 1203.29    | \$ 306,363.64 | Sold   | C0037       | Individual | Tyler      | Carr     | 32                      | 26-35    | 1974 | M      | USA     | California | Home       | 5 Yes             |          | Client  |   |    |
| 32 | 3033                                   | 3        | 2007         |               | 9 Apartment      | 33         | 670.89     | \$ 200,300.63 | Sold   | C0127       | Individual | Maia       | Chandler | 32                      | 26-35    | 1975 | F      | USA     | Utah       | Home       | 1 Yes             |          | Website |   |    |
| 33 | 1025                                   | 1        | 2005         |               | 3 Apartment      | 25         | 1434.09    | \$ 382,041.13 | Sold   | C0018       | Individual | Rodrigo    | Ramirez  | 33                      | 26-35    | 1972 | M      | USA     | California | Home       | 1 No              |          | Agency  |   |    |
| 34 | 3027                                   | 3        | 2006         |               | 8 Apartment      | 27         | 781.07     | \$ 245,572.79 | Sold   | C0040       | Individual | Gianni     | Fritz    | 33                      | 26-35    | 1973 | M      | USA     | California | Home       | 3 No              |          | Website |   |    |
| 35 | 3031                                   | 3        | 2007         |               | 3 Apartment      | 31         | 1596.35    | \$ 407,214.29 | Sold   | C0080       | Individual | Janelle    | Espinoza | 33                      | 26-35    | 1974 | F      | USA     | California | Investment | 4 No              |          | Client  |   |    |
| 36 | 2043                                   | 2        | 2007         |               | 4 Apartment      | 43         | 1110.32    | \$ 355,073.40 | Sold   | C0083       | Individual | Julio      | Leonard  | 33                      | 26-35    | 1974 | M      | USA     | Nevada     | Home       | 2 Yes             |          | Website |   |    |
| 37 | 3034                                   | 3        | 2007         |               | 4 Apartment      | 34         | 781.07     | \$ 256,821.64 | Sold   | C0085       | Individual | Melanie    | Holland  | 33                      | 26-35    | 1974 | F      | USA     | California | Investment | 3 No              |          | Client  |   |    |
| 38 | 3016                                   | 3        | 2007         |               | 4 Apartment      | 16         | 697.89     | \$ 226,342.80 | Sold   | C0091       | Individual | Gordon     | Brown    | 33                      | 26-35    | 1974 | M      | UK      | Arizona    | Investment | 4 No              |          | Agency  |   |    |
| 39 | 1018                                   | 1        | 2004         |               | 10 Apartment     | 18         | 625.80     | \$ 191,389.87 | Sold   | C0007       | Individual | Jaelyn     | Berger   | 34                      | 26-35    | 1970 | F      | USA     | California | Home       | 3 No              |          | Website |   |    |
| 40 | 2050                                   | 2        | 2006         |               | 9 Apartment      | 50         | 957.53     | \$ 297,008.97 | Sold   | C0048       | Individual | Adriana    | Shaffer  | 34                      | 26-35    | 1972 | F      | USA     | California | Investment | 2 No              |          | Agency  |   |    |
| 41 | 2044                                   | 2        | 2007         |               | 1 Apartment      | 44         | 722.96     | \$ 250,773.15 | Sold   | C0065       | Individual | Mara       | Franco   | 34                      | 26-35    | 1973 | F      | USA     | California | Home       | 5 Yes             |          | Agency  |   |    |
| 42 | 3039                                   | 3        | 2007         |               | 5 Apartment      | 39         | 923.21     | \$ 312,211.14 | Sold   | C0096       | Individual | Kael       | Hurley   | 34                      | 26-35    | 1973 | M      | USA     | Colorado   | Home       | 3 No              |          | Website |   |    |
| 43 | 3053                                   | 3        | 2007         |               | 12 Apartment     | 53         | 670.24     | \$ 190,119.50 | Sold   | C0164       | Individual | Belinda    | Hogan    | 34                      | 26-35    | 1973 | F      | USA     | California | Home       | 1 Yes             |          | Agency  |   |    |

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365 DataScience RE California Database

Gender

Frequency distribution table

|        |     |  | Frequency | Relative frequency |
|--------|-----|--|-----------|--------------------|
| Male   | M   |  | 108       | 55%                |
| Female | F   |  | 70        | 36%                |
| Firms  | N/A |  | 17        | 9%                 |
| Total  |     |  | 195       | 100%               |

A 3D pie chart illustrating the gender distribution of the data. The chart is divided into three segments: a large red segment representing Male at 55%, a medium green segment representing Female at 36%, and a small dark blue segment representing Firms at 9%. The segments are labeled with their respective percentages in white text. A legend below the chart identifies the colors: red for Male, green for Female, and dark blue for Firms.

| Gender | Frequency | Relative frequency |
|--------|-----------|--------------------|
| Male   | 108       | 55%                |
| Female | 70        | 36%                |
| Firms  | 17        | 9%                 |

N.B. Firms have no gender. However, we need to add them to this pie chart, as otherwise, we will get a wrong interpretation of the data.

C10       $\times$      $\checkmark$      $f_x$     =COUNTIF('365RE'!\$W\$6:\$W\$272,B10)

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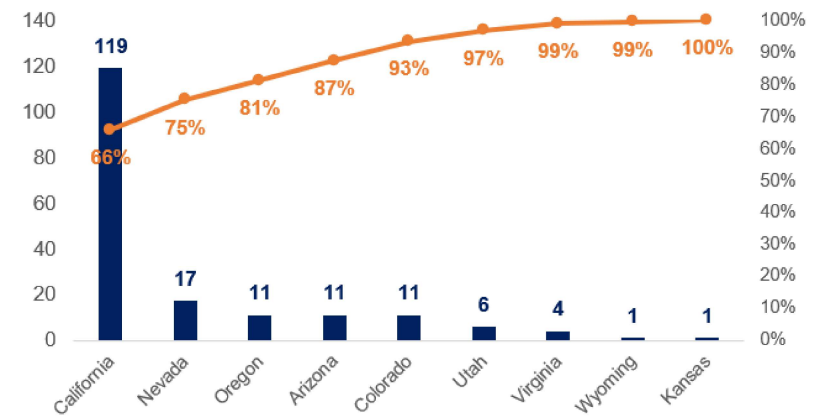
1    **365 DataScience RE California Database**

2    **Location**

4    **Frequency distribution table**

|                     | Frequency | Relative frequency | Cumulative frequency | Cumulative US only |
|---------------------|-----------|--------------------|----------------------|--------------------|
| 7    California     | 119       | 45%                | 45%                  | 66%                |
| 8    Nevada         | 17        | 6%                 | 51%                  | 75%                |
| 9    Oregon         | 11        | 4%                 | 55%                  | 81%                |
| 10    Arizona       | 11        | 4%                 | 59%                  | 87%                |
| 11    Colorado      | 11        | 4%                 | 63%                  | 93%                |
| 12    Utah          | 6         | 2%                 | 66%                  | 97%                |
| 13    Virginia      | 4         | 1%                 | 67%                  | 99%                |
| 14    Wyoming       | 1         | 0%                 | 67%                  | 99%                |
| 15    Kansas        | 1         | 0%                 | 68%                  | 100%               |
| 16    None (abroad) | 86        | 32%                | 100%                 |                    |
| 17    Total         | 267       | 100%               |                      |                    |

**Segmentation of US clients by State**



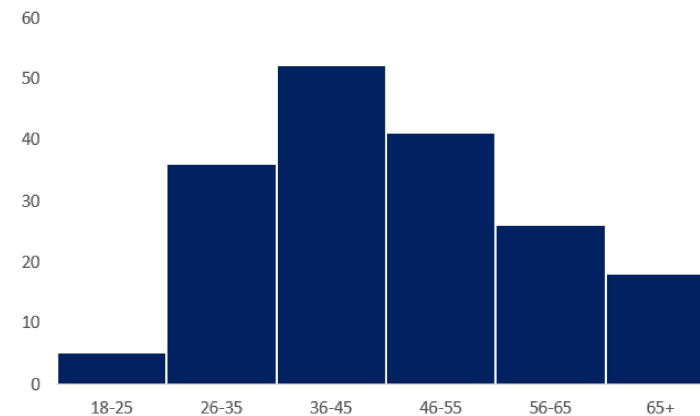
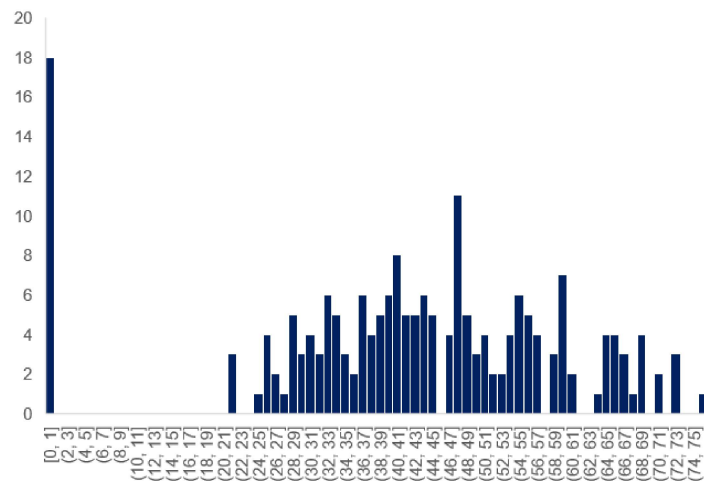
21    This is a Pareto diagram. The bars show the absolute frequency of each category. In this case this translates to: how many customers are there from each state. For example, there are 4 customers from Virginia.  
22    The secondary axis (the orange line), shows the cumulative frequency. For example, California, Nevada, and Oregon together, account for 81% of all purchases.

27    N.B. Clients that are not from the USA don't have a state value. In the table, they are listed as 'None'. We need to add them to the frequency distribution table, as otherwise, we will get a wrong interpretation of the data.  
28    However, as the Pareto diagram represents clients from the USA, we don't have to include 'None' there.

**Age**

### Frequency distribution table

|          |        |
|----------|--------|
| Mean     | 46.15  |
| Median   | 45.00  |
| Mode     | 48.00  |
| Skew     | 0.24   |
| Variance | 164.91 |
| St. dev. | 12.84  |



The first graph represents the frequency distribution of the variable age. We have a high bar at 0 as that is the null values associated with corporate clients.

The second graph is a histogram, based on the intervals from the original table. It is built using the data from the frequency distribution table above. Firms are not included in it.

Note: if you reorder the Spreadsheet 365RE some values may change

